

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Mon, Jul 13 · 6:00 AM – 10:00 AM | PDOP 0.99 · 34 sv | NEAREST CORS
P261 · 14.1 mi ✓

REPORT CONTENTS

01	7-DAY GNSS FIELD RANKING	202	SITE & SIGNAL REPORT	3
03	BEST STATIC OBS WINDOW	404	EMERGENCY HOSPITALS	5
05	DAY-BY-DAY WEATHER + DRONE	606	AI FIELD ANALYSIS	13
07	WATER LEVELS & GAUGES	1408	FAULTS & SUBSIDENCE	15
09	STRUCTURAL PARAMETERS	1610	SOILS & ENGINEERING	16
11	MINES & MINERALS	1712	NATIVE PLANTS & WILDLIFE	18
13	FCC MICROWAVE PATHS	1914	SOLAR & PASSIVE DESIGN	21
15	AI SITE INTELLIGENCE BRIEF	2316	FIELD MAP · TOPO · DXF SITE PLAN	24



Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com



This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

16.6 TECU

-5.6 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork

✖ GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast

9 mph

GO

High 93°F

Low 59°F

Wind max 9 mph

Gusts 12 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO - GOOD CONDITIONS

Partly cloudy | Wind: 9.4 mph (gusts 11.6 mph) | Precip: 0" | Temp: 59°–93.2°F



DRONE WIND ANALYSIS



GO Peak: 9.4/11.6 mph S

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	2/4 mph		SE
6 AM	0/3 mph		W
7 AM	3/5 mph		E
8 AM	2/2 mph		SSW
9 AM	2/3 mph		SE
10 AM	4/5 mph		SSE
11 AM	5/5 mph		S
12 PM	5/6 mph		S
1 PM	5/5 mph		S
2 PM	9/10 mph		S
3 PM	7/8 mph		S
4 PM	8/9 mph		SSW
5 PM	9/11 mph		SSW
6 PM	9/11 mph		SSW
7 PM	9/12 mph		S
8 PM	8/11 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

GNSS SURVEY OPERATIONS — NAPA COUNTY, CA

Report Date: Monday, July 13, 2026 | Survey Start: Monday, July 13, 2026

VERDICT: GO

Excellent PDOP geometry (0.94–0.99), favorable weather, and moderate ionospheric conditions support survey execution today. Proceed with standard protocols.

BEST SURVEY WINDOW

Monday, July 13, 2026: 06:00 AM – 10:00 AM (Pacific Daylight Time)

PDOP 0.94 with 34 satellites (multi-constellation, 15° elevation mask). This 4-hour window offers optimal geometric strength. Morning conditions minimize atmospheric heating and ionospheric delay variance. Recommend prioritizing critical control and kinematic work within this window.

BEST 4-HOUR STATIC GNSS WINDOW

Monday, July 13, 2026: 06:00 AM – 10:00 AM

Deploy static receivers at primary control points during this interval. Constellation stability and PDOP of 0.94 will deliver sub-centimeter repeatability. Moderate Kp (peak 4.7, avg 2.4) is manageable with standard 4-hour static occupation. TEC estimate is degraded but shows -5.6 TECU below 30-day median, indicating favorable ionospheric conditions relative to site history. If extended occupation (8+ hours) is required, initiate by 06:00 AM to maximize daylight and reduce evening ionospheric variability.

DRONE OPERATIONS ASSESSMENT

Weather is marginal for drone deployment. Overcast skies and peak winds to 9 mph are within operating limits for most survey platforms. No precipitation forecast. Visual line-of-sight operations are feasible in daylight hours (06:00 AM – sunset, ~20:30 hours). RTK/PPK photogrammetry recommended to avoid reliance on single-frequency corrections during moderate ionospheric activity. Confirm drone battery performance in 93°F ambient temperature.

FIELD SAFETY WARNINGS

— GPS PRN 11 is unhealthy (UNUSUFN); all modern receivers auto-exclude. No field action required.

— No active wildfires within reporting range. Air quality and visibility adequate.

— Terrain elevation mask 13° based on dwarf scrub cover (low sky obstruction, low multipath). Standard 15° elevation mask is appropriate; terrain will not degrade solution.

— Moderate geomagnetic activity (Kp 4.7) may increase ionospheric delay variability.
Standard mitigation: longer static occupations and real-time monitoring of dilution
of precision during kinematic operations.

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SUMMARY

Survey conditions are favorable for Monday, July 13, 2026. PDOP geometry is excellent (0.94 best case), weather is stable with no precipitation, and ionospheric activity is moderate and manageable with standard field protocols. The 06:00–10:00 AM window is optimal for static and kinematic positioning. Deploy primary control and high-accuracy work within this interval. Drone operations are feasible in daylight with attention to wind and thermal effects. GPS satellite constellation health is good (one unhealthy satellite automatically excluded; no impact on PDOP calculation). Proceed with survey as planned.

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Report prepared by: GNSS Survey Operations Planning System
Data sources: NAVCEN, NOAA Space Weather Prediction Center, GNSS Planning Online,
USGS NLCD, National Interagency Fire Center

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

Download .txt File

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.03 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.93 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.16 ft (-18 ft)	466 ft	787K af	81%
Lake Oroville	93.4 mi	861.09 ft (-39 ft)	900 ft	2,857K af	81%
New Melones Lake	100.1 mi	1,020.15 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

○ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa County Coastal Alluvium and Sandstone-Limestone (N100CACSTL) is a shallow to moderate-depth aquifer composed of unconsolidated alluvial deposits and sedimentary bedrock formed during Quaternary and Tertiary time. Water moves through sand and gravel layers and fractured sandstone, recharged primarily by local precipitation and streamflow infiltration in the Napa Valley. At this site, water is readily accessible at depths ranging from 10 to 88 feet, making it suitable for moderate-capacity wells.

FORMATION

Name	Napa County Coastal Alluvium and Sandstone-Limestone
Age / Rock	Quaternary to Tertiary age; unconsolidated alluvium, sandstone, and limestone.
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	9–295 ft
Typical Depth	50–120 ft
Recharge Source	Precipitation on valley floor and hillside outcrops; streamflow infiltration along Napa River and tributaries

Transit Time	5–25 years to 100 ft depth (est. USGS/DWR)
WATER QUALITY	
pH	est. 6.8–7.6 (est. USGS/DWR)
Hardness	est. 180–320 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–500 mg/L (est. USGS/DWR)
Notes	No sample data available; typical for formation — expect moderate hardness and neutral pH suitable for domestic and agricultural use.
TREND	
Trend	Water levels show decline from 1977 (9.90–54.10 ft BGS) to recent measurements (21.86–87.71 ft BGS), indicating net drawdown of 15–65 ft over 45 years consistent with regional pumping.
Source	Site observation data (1977–2022)
Seasonal Swing	est. 8–20 ft (est. USGS/DWR regional)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU	
The Coast Ranges near 38.3459, -122.3047 are characterized by consolidated bedrock and well-drained soils with limited clay sequences, making them geologically unsuitable for significant land subsidence. The area lies outside critically overdrafted groundwater basins, and the N100CACSTL aquifer system does not exhibit the conditions necessary for compaction-driven subsidence—namely, thick clay interbeds subject to prolonged groundwater withdrawal. For surveyors working in this region, elevations can be treated as stable, though standard benchmark verification protocols should remain part of routine practice.	
SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<0.5 mm/yr (est.)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN	
This marine climate features mild temperatures year-round with moderate rainfall, so your primary envelope concern is moisture management and air sealing rather than extreme temperature swings—prioritize quality water barriers and continuous air gaps in wall assemblies. Maximize south-facing glazing to capture passive solar gains during cooler months, but provide deep overhangs or exterior shading on west-facing windows to prevent summer heat gain from the afternoon sun, and minimize north-facing glass since it provides little benefit in this temperate zone. The most effective passive strategy here is natural ventilation through operable windows aligned for cross-breezes, which can reduce cooling loads significantly during the mild shoulder seasons and evenings.	
🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU	
This coastal marine site in Sonoma County presents moderate wind exposure and light snow loading as the primary design drivers. Seismic parameters are currently unavailable; a geotechnical investigation and seismic hazard assessment are anticipated to establish appropriate foundation and lateral load design criteria. The well-drained gravelly loam and absence of subsidence risk suggest conventional shallow foundations may be viable, pending soil confirmation.	
FOUNDATION	
Type	Conventional spread footing anticipated pending geotechnical investigation

Basis Well-drained gravelly loam with favorable drainage characteristics suggests shallow foundation may be adequate; however, seismic site classification (SDC) is unknown and may warrant deeper investigation or special detailing—consult a licensed engineer

STRUCTURAL IMPLICATIONS

- Seismic** Seismic design category cannot be determined from available data (Ss, S1, SDS, SD1 all null). A seismic hazard assessment and site-specific response analysis may be warranted before finalizing lateral load-resisting system. Site Class D is noted for planning; confirm with geotechnical engineer.
- Wind Exposure** Exposure C (open terrain with scattered obstructions) suggested per ASCE 7-22 for this coastal location—verify site-specific topography and surrounding conditions
- Wind** 92 mph 3-second gust wind speed (Risk Category II) anticipated—standard wind load design likely adequate; no hurricane or extreme wind exposure anticipated for this location
- Snow** Light snow load of 5.8 psf anticipated—standard roof design and drainage provisions likely adequate; winter wind factor (W2=0.35) may suggest modest snow drift considerations in exposed areas

INVESTIGATIONS & INSPECTIONS

- Geotech Required** Yes—geotechnical investigation recommended to confirm soil bearing capacity, establish seismic parameters (SDC, site class, liquefaction screening), and validate frost depth estimate
- Liquefaction Screen** Screening recommended—Site Class D and proximity to coastal zone may warrant liquefaction potential assessment; confirm groundwater depth and soil composition with site investigation
- Special Inspections** May be required once seismic design category is confirmed; not anticipated to be needed for wind or snow alone—consult engineer after seismic parameters are established
- Code Edition** ASCE 7-22 / IBC 2021 (planning reference only)
- Site Class Note** Site Class D assumed for planning purposes; geotechnical investigation strongly recommended to confirm soil properties and refine seismic site classification before design finalization

SOILS — USDA SOIL SURVEY

- MAP UNIT:** Coombs gravelly loam, 2 to 5 percent slopes **SERIES:** Coombs
- DRAINAGE CLASS:** Well drained **HYDRO GROUP:** C
- SOIL ORDER:** Alfisols **SUBORDER:** Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in alluvial or colluvial material derived from mixed rock sources in the Coast Ranges of California. The gravelly surface layer overlies a clayey argillic subsoil, creating moderately good bearing capacity but requiring attention to moisture conditions during fill placement. Walking onto this site, you will encounter gentle slopes with coarse surface materials that transition to finer, more cohesive subsurface layers—typical behavior for foothill alluvial soils in Mediterranean climates.

Mixed alluvial/colluvial parent material on 2–5% slopes; historically grassland and oak savanna vegetation.

DRAINAGE

- Class** Well drained
- Seasonal Saturation** None within 60 inches
- Flooding Risk** Low — gentle slopes away from active drainages
- Ponding Risk** None

ENGINEERING

- Bearing Capacity** 2,500–3,500 psf (shallow foundations, gravel-stable surface)
- Rating** Good
- Shrink-Swell** Moderate — clay-rich argillic subsoil (B horizon)
- Frost Action** Low — favorable for foundations in freeze-thaw zones
- Cut Suitability** Excellent (gravel-rich A and upper B horizons provide stable material)
- Fill Suitability** Fair to Good (clay subsoil acceptable if moisture managed; compact near optimum water content)

SEPTIC

- Suitability** Conditional
- Notes** Perc test required — clay loam subsoil permeability may be marginal; 12–24 inch vertical separation to seasonal water table; surface gravels promote lateral drainage.

HAZARDS

- Excavation** Moderate — 25–40% gravel and cobble content may slow equipment; rippable without blasting in most conditions
- Expansive Clay** Low to Moderate — clay content 15–25%; monitor cracks in fill or foundation designs sensitive to movement
- Erosion** Low on in-situ slopes; Moderate after disturbance — gravel armoring lost; apply erosion control on cuts and fills
- Other** Subsurface clay layer may trap shallow perched water during wet season; confirm drainage design for cut slopes over 6 feet

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USBR RISE · [data.usbr.gov](#) · USACE CWMS · [water.usace.army.mil](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 13, 2026, 3:32 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, well beyond any distance threshold requiring foundation, environmental, or surface rights review. No active producers, injection wells, or idle wells exist within actionable range.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200
total in search area

10
within 5 mi

108
within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active producers operate within 5 miles of the site. Basalt Rock Co (2 mi, clay operation) and Mt George Pumice Deposit (3.1 mi, pumice producer) will generate heavy truck traffic and dust during operating hours, while Silva Pumice Quarry (5 mi, pumice producer) presents similar concerns at the periphery. Blasting and crushing noise should be anticipated from the pumice operations.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6
Endangered

6
Threatened

12
Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	Ambystoma californiense	Endangered

COMMON NAME	SCIENTIFIC NAME	STATUS
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Assessment – 38.3459, -122.3047 **Section 7 Consultation Triggers:** Federal nexus (permits, funding, or approvals) would trigger mandatory Section 7 consultation given 12 species with critical habitat within 25 miles. **Most Planning-Relevant Species:** California tiger salamander, vernal pool tadpole shrimp, and California red-legged frog present the highest regulatory risk due to their dependence on seasonal wetlands and vernal pools, which are common landscape features in the North Bay region and frequently impacted by development. **Recommended Next Step:** Conduct a jurisdictional delineation of Waters of the U.S. and a focused biological survey during appropriate seasonal windows to map vernal pools, seasonal wetlands, and upland habitat; results will determine whether formal habitat assessment and incidental take authorization are required. Early coordination with USFWS and CDFW is advisable to clarify applicability and streamline permitting timelines. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15

COMMON NAME	SCIENTIFIC NAME	LAST OBS
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smaliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site occurs within a **mixed oak woodland and chaparral community**, dominated by drought-tolerant species like blue oak, California black oak, and California buckeye, with understory elements including aromatic shrubs (California mugwort, California bay) and spring ephemeral wildflowers. The prevalence of xerophytic plants—particularly clay mariposa lily, canyon liveforever, and cliffbrake ferns—indicates **well-drained, seasonally dry soils typical of foothill or upland terraces** with limited water retention. Grading and drainage work should anticipate **rapid runoff during winter rains and minimal summer moisture availability**, requiring careful erosion control measures and consideration of the shallow, rocky soils common to oak woodlands; excessive drainage modification could destabilize native plant communities adapted to natural moisture cycles. The species diversity and abundance of observations suggest an ecologically intact site with significant restoration or conservation value. **Consult a licensed botanist for site-specific vegetation assessment.**

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

636	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		636 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		1100 obs	
Datura wrightii		57 obs	

SPECIES	OBSERVATIONS
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Northern California Survey Site **Highest-Priority Hazards:** Your greatest risks are encounters with Northern Pacific rattlesnakes (636 documented observations) and poison oak, which dominates the area with 1,439 plant observations. Both are common in Sonoma County survey terrain and cause the majority of field injuries in this region. **Practical Avoidance:** Wear closed-toe boots, long pants, and gaiters—rattlesnakes strike at ankle height and are most active in warm months. Learn to recognize poison oak's three-leaflet clusters and avoid contact; wash exposed skin and clothing immediately after work. Watch for widow spiders in dark shelters (equipment, stored materials) before handling them. Stay alert on trails, especially near brush and rock outcrops where snakes shelter. **Seasonal Considerations:** Rattlesnake activity peaks April–October when temperatures warm; exercise heightened caution during these months. Poison oak remains a hazard year-round but causes more dermatitis in spring and summer when the plant is actively toxic. Stinging insects (velvet ants and wasps) are most aggressive July–September. **Always verify current conditions locally before fieldwork.**

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200 89 64 5 14 6
Total Observations Total Species Birds Mammals Reptiles Amphibians

BIRDS (64 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	8
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melozone crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Northern Flicker	<i>Colaptes auratus</i>	3
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltriparus minimus</i>	3

+ 49 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Coyote	<i>Canis latrans</i>	3
Black-tailed Jackrabbit	<i>Lepus californicus</i>	3
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	7
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1

COMMON NAME	Scientific Name	OBS
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
Western Toad	<i>Anaxyrus boreas</i>	3
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Rough-skinned Newt	<i>Taricha granulosa</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment The site at 38.3459, -122.3047 (northern California wine country) supports a diverse wildlife community characteristic of oak woodland and grassland habitat with riparian influences, evidenced by the co-occurrence of upland species (California Quail, Western Bluebird, California Towhee) and water-dependent species (egrets, pond turtles, chorus frogs). The presence of sensitive species including Foothill Yellow-legged Frog, California Giant Salamander, and Northwestern Pond Turtle indicates proximity to seasonal or perennial streams with intact riparian vegetation and cool-water refugia; these species and the diverse native amphibian and reptile community suggest relatively low chemical disturbance and adequate aquatic habitat connectivity. The mammalian assemblage (native Mule Deer, Coyote, Jackrabbit alongside introduced Eastern Fox Squirrel) and predatory bird diversity (raptors, egrets) indicate a moderately functional food web, though presence of American Bullfrog and potential grazing pressure suggest some habitat modification from baseline conditions. Overall, this site retains significant ecological value as wildlife habitat with moderate sensitivity to disturbance, particularly for aquatic and riparian-dependent species. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↔ excellent
7/12	30.3°	98.8%	↔ excellent
8/12	33.7°	99.3%	↔ excellent
9/12	36.9°	99.8%	↔ annual optimal
10/12	39.8°	99.8%	↔ excellent
12/12	45°	99%	↔ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🏠 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a great spot for solar—you get about 6.6 peak sun hours most days, which is like having a reliable 9-to-5 job for your panels, and that jumps to 9 hours in summer when you need cooling power most. Your roof can face south at a 38-degree angle (roughly a 9-to-12 pitch) to catch that sun perfectly, but here's the thing: if your roof is dark, it's absorbing so much summer heat that it's like leaving a black car in the sun—painting it light colors could cut that heat gain by two-thirds. West-facing windows are your trickiest problem because they get hammered by afternoon sun (especially in summer), so that's where you'll want good shading, heavy curtains, or reflective film to keep things cool. A big deciduous tree planted on the south side of your house is actually clever planning—it'll block about 4 hours of intense summer sun while still letting winter sun through when the leaves drop, giving you free shade when you need it most. Thermal mass just means that materials like concrete or brick in your walls slow down temperature swings, so your home doesn't heat up or cool down like a tin can; thicker construction (like ICF walls) holds heat nearly 8 times longer than standard framing. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING 38.34586°N, 122.30466°W | Napa County, California

Structural Design Drivers & Climate Envelope

This site falls within IECC Climate Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, indicating no meaningful frost cycling and thus no freeze-thaw loading on exposed elements or foundations. Wind and snow data are not available for this location; however, the marine classification and zero AFI together suggest a coastal maritime influence with mild winters and modest seasonal loading. Without quantified wind or snow parameters, seismic loading emerges as the likely controlling design driver for a site in Napa County—a tectonically active region—until wind speed and snow load data specific to this coordinate can be obtained from ASCE 7 maps or the California Building Standards Commission. The architect should confirm current design wind speed and roof snow load for this exact latitude and longitude before finalizing structural system type.

Geotechnical Foundation Design

Soils on site are mapped as Coombs gravelly loam, 2 to 5 percent slopes, with Hydrologic Group C designation and well-drained drainage class. Group C soils have moderate infiltration rates and moderate runoff potential; "well drained" status indicates no perched water table or saturation issues expected under normal conditions. The gravelly loam composition and slope grade favor conventional shallow foundations (spread footings or mat) without need for special dewatering or underdrain systems. Frost depth is not provided for this location, but the zero AFI makes frost heave a non-issue; footing depth should be determined by bearing capacity and slope stability rather than frost protection. A geotechnical site investigation with field borings should confirm bearing capacity (CBR or SPT N-values) and verify the absence of subsurface fill or voids, particularly given proximate mining activity.

Seismic & Fault Assessment

No Quaternary faults are recorded within the search radius of this coordinate, which substantially reduces the immediate surface rupture hazard. However, Napa County lies within a seismically active region; the absence of mapped Quaternary faults does not eliminate background seismic shaking risk. No subsidence basin data is on record, eliminating subsidence as a primary concern. A Fault Rupture Hazard Zone map search and site-specific seismic hazard assessment (per California Geological Survey guidelines) are recommended to establish design spectral accelerations and confirm the applicability of standard ASCE 7 seismic design categories. Wildfire risk data are not provided; consultation with local fire marshals and Cal Fire regarding defensible space and ignition-resistant material requirements should proceed in parallel with structural design.

Mining & Site Context

Three active mineral producers operate within 5 miles of this site: Basalt Rock Co. (2 miles, clay), Mt George Pumice Deposit (3.1 miles, pumice), and Silva Pumice Quarry (5 miles, pumice). The proximity of these quarries warrants confirmation that extraction activities and blast zones do not pose a ground vibration or flyrock hazard to the proposed structure. Approximately 200 MRDS deposits exist within a ~35-mile radius, typical of this geologically productive region; the nearest operating facilities are aggregate and industrial mineral producers rather than metals mining, reducing acid mine drainage concerns. An inquiry with each operator regarding blast schedules, vibration monitoring protocols, and any reciprocal impact agreements should be conducted during the due diligence phase.

Bottom Line

Obtain site-specific seismic design parameters (ASCE 7 mapped values or Site Class assessment) and quantified wind/snow loads before selecting the lateral load-resisting system; conduct a geotechnical boring program to verify bearing capacity and rule out subsurface voids related to nearby mining activity.

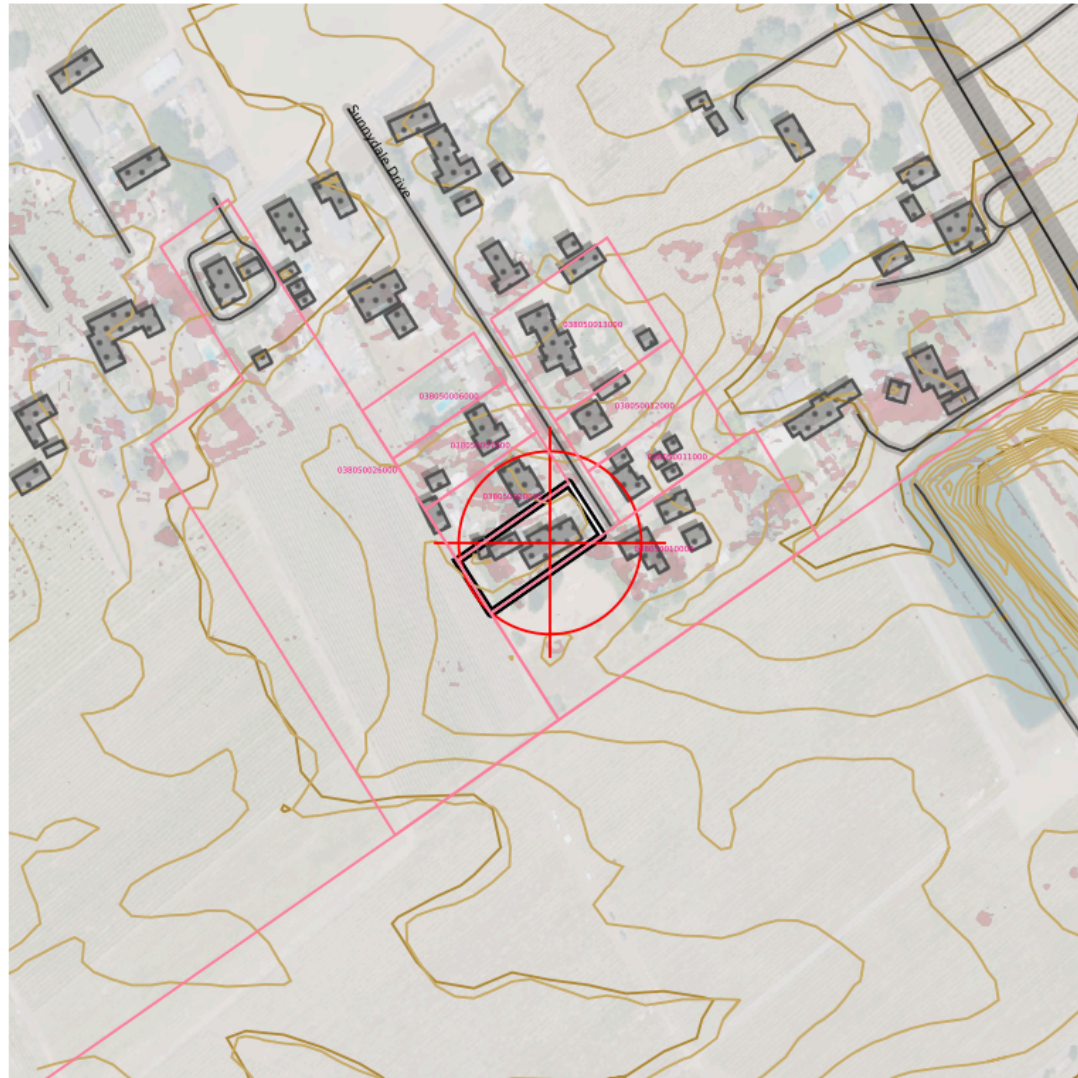


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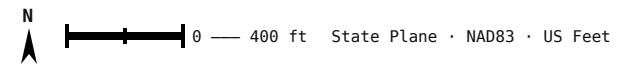
Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.



TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



- Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
- County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polyline (all geometry)



SITE AERIAL – 600 ft × 600 ft

38.34586°N, 122.30466°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

38.34586°N, 122.30466°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

